Practice workbook · Writing the equation of a line through two points

Finding the gradient from two points (or from a gradient and one point) and writing the straight-line equation in the form the question asks for.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-03

1. Question 8 · part a · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part a only.

8 The points A and B have coordinates $(-6, 8)$ and $(12, 2)$ respectively.

(a) Find an equation of the straight line passing through A and B in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

(b) Find the exact length of AB

(2)

The point X with coordinates (m, n) lies on AB such that $AX:XB = 1:2$

(c) Find the value of m and the value of n

(2)

The line L passes through the point X and is perpendicular to AB
The point C with coordinates (p, q) lies on L where $p > 0$ and $q > 0$

Given that AB is a diameter of a circle and C also lies on the circumference of the circle,

(d) find

(i) the exact value of p

(ii) the exact value of q

(7)

(e) Find the exact area of triangle ABC

(3)

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1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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2. Question 11 · part a · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

11 The points A and B have coordinates $(-1, 3)$ and $(5, 6)$ respectively.

(a) Find an equation for the line AB.

(2)

The point P divides AB in the ratio $2 : 1$

(b) Show that the coordinates of P are $(3, 5)$

(2)

The point C with coordinates (m, n), where $m > 0$, is such that CP is perpendicular to the line AB.

Given that the radius of the circle which passes through A, P and C is 5

(c) find the value of m and the value of n.

(6)

The point D with coordinates (p, q) is such that the line AD is perpendicular to the line AB and the line DC is parallel to the line AB.

(d) Find the value of p and the value of q.

(3)

(e) Find the area of trapezium $ABCD$.

(4)

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2. Question 11 · part a · 2 marks · Working and answer

Working and answer

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3. Question 2 · part a · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

2 The point A has coordinates $(-7, -1)$ and the point B has coordinates $(3, 4)$

(a) Find an equation of the line that passes through A and B
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(3)

The point C has coordinates $(-3, 7)$

Given that k is a constant such that $AB = kAC$

(b) find the value of k

(2)

The point D has coordinates $(3, p)$ where p is a constant.

Given that CD is perpendicular to AB

(c) find the value of p

(3)

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3. Question 2 · part a · 3 marks · Working and answer

Working and answer

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4. Question 5 · part b · 2 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 The line l with gradient $-\frac{1}{12}$ passes through the points A and B with coordinates $(p, 10)$ and $(123, 0)$ respectively.

(a) Show that $p = 3$

(2)

(b) Find an equation for l in the form $rx + sy + t = 0$ where r, s and t are integers.

(2)

The line k is perpendicular to l and passes through the point A.

(c) Find an equation for k in the form $y = mx + c$

(3)

Line k intersects the x-axis at the point C.

(d) Find the exact area of triangle ABC.

(4)

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4. Question 5 · part b · 2 marks · Working and answer

Working and answer

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5. Question 9 · part a · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9 The point A has coordinates $(-4, 3)$ and the point B has coordinates $(6, 8)$
The points A and B lie on the line k

(a) Find an equation of k

(2)

The point C, on k, is such that $AC : CB = 4:1$

(b) Find the coordinates of point C

(2)

The point D with coordinates (p, q), where $p < 0$, lies on the line l through C that is perpendicular to k

The length of CD is $8\sqrt{5}$

(c) Find the coordinates of D

(6)

(d) Find the area of triangle ACD

(2)

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5. Question 9 · part a · 2 marks · Working and answer

Working and answer

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6. Question 4 · part a · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

4 The point A has coordinates $(-4, -10)$ and the point B has coordinates $(3, 11)$
The line l passes through A and B.

(a) Find an equation of l.

(2)

The point P lies on l such that $AP : PB = 3 : 4$

(b) Find the coordinates of P.

(2)

The point Q with coordinates (m, n), where $m < 0$, lies on the line through P that is perpendicular to l.

Given that the length of PQ is $\sqrt{10}$

(c) find the coordinates of Q.

(6)

The point R has coordinates $(-11, -21)$

(d) Show that

(i) AB and RQ are equal in length,

(ii) AB and RQ are parallel.

(4)

(e) Find the area of the quadrilateral $ABQR$.

(2)

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6. Question 4 · part a · 2 marks · Working and answer

Working and answer

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7. Question 2 · part a.i · 0 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

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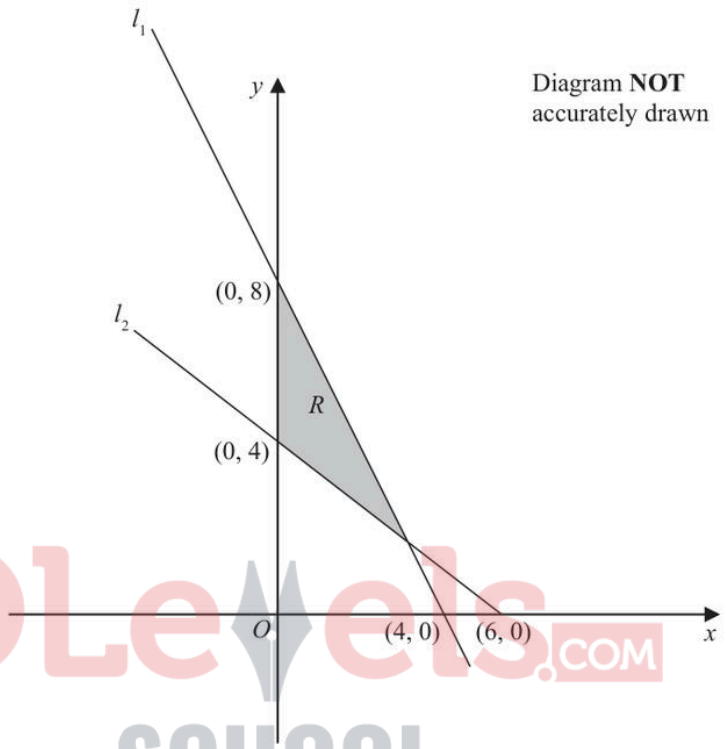


Figure 2

Figure 2 shows the shaded region R bounded by the line l_1, the line l_2 and the y-axis.

The points with coordinates (0, 8) and (4, 0) lie on l_1

The points with coordinates (0, 4) and (6, 0) lie on l_2

(a) Find, in the form $ax + by = c$, where a, b and c integers, an equation of

(i) l_1

(ii) l_2

(3)

(b) Hence write down three inequalities that define the region R.

(3)

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7. Question 2 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 2 · part a.ii · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

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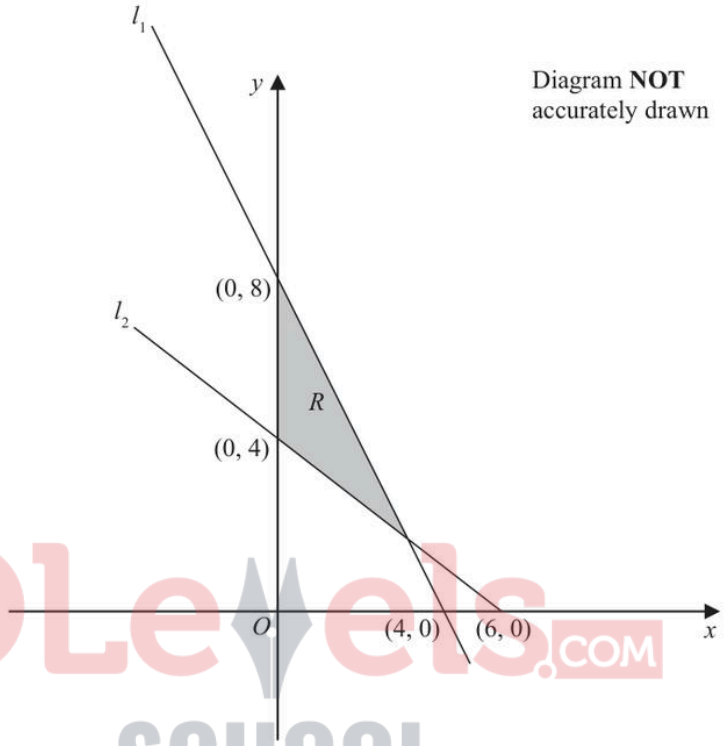


Figure 2

Figure 2 shows the shaded region R bounded by the line l_1, the line l_2 and the y-axis.

The points with coordinates (0, 8) and (4, 0) lie on l_1

The points with coordinates (0, 4) and (6, 0) lie on l_2

(a) Find, in the form $ax + by = c$, where a, b and c integers, an equation of

(i) l_1

(ii) l_2

(3)

(b) Hence write down three inequalities that define the region R.

(3)

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8. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

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9. Question 8 · part a · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 The points A and B have coordinates $(1, 5)$ and $(9, 9)$ respectively.

(a) Find an equation of line AB, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

The line l is perpendicular to AB and passes through the point X which lies on AB such that $AX : XB = 3:1$

(b) Show that an equation of l is $y = -2x + 22$

(5)

The point C has coordinates $(6, p)$

Given that C lies on l

(c) find the value of p

(1)

$ABCD$ is a parallelogram where the x coordinate of D is negative.

(d) Find the coordinates of the point D

(3)

(e) Find the area of the parallelogram $ABCD$

(4)

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9. Question 8 · part a · 3 marks · Working and answer

Working and answer

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10. Question 11 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

11 The points A and B have coordinates $(-3, -5)$ and $(7, 5)$ respectively.

(a) Find an equation for the line AB

(2)

The point C has coordinates $(p, 1)$ where $p < 0$

Given that AC and BC are perpendicular,

(b) prove that $p = -5$

(7)

The point D, where BCD is a straight line, is such that C divides BD in the ratio $4:3$

(c) Find the coordinates of D

(2)

(d) (i) Find the exact length of AC

(ii) Hence, or otherwise, find the area of triangle ABD

(4)

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10. Question 11 · part a · 2 marks · Working and answer

Working and answer

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11. Question 2 · part b · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

2 The point A has coordinates $(3, 2)$, the point B has coordinates $(8, 3)$ and the point C has coordinates $(4, 7)$

(a) Show that ABC is an isosceles triangle.

(2)

The midpoint of BC is M

(b) Find an equation of the line that passes through A and M

Give your answer in the form $y = mx + c$

(3)

The points A, C and D are collinear such that $AD = kAC \quad (k > 1)$

Given that $\angle ABD = 90^\circ$

(c) find the coordinates of D
Show your working clearly.

(8)

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11. Question 2 · part b · 3 marks · Working and answer

Working and answer

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12. Question 10 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.

10 The equation of the line L_1 is $y - 3x + a = 0$

The point A with coordinates $(a, 10)$ lies on L_1

(a) Show that $a = 5$

(2)

Line L_1 crosses the y-axis at the point B

(b) Write down the coordinates of B

(1)

The point C with coordinates $(50, -5)$ lies on L_2

Given that L_2 passes through A

(c) (i) show that L_1 and L_2 are perpendicular

(3)

(ii) hence find an equation for L_2 giving your answer in the form $y = px + q$

(2)

The point D has coordinates (m, n)

The length of CD is $15\sqrt{10}$ and the gradient of BD is -3

(d) Find the value of m and the value of n

(6)

(e) Find the area of quadrilateral $ABCD$

(3)

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12. Question 10 · part c.ii · 2 marks · Working and answer

Working and answer

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